

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce Department of Information Technology (B.Sc.I.T
Semester IV)
MERN

Practical – II

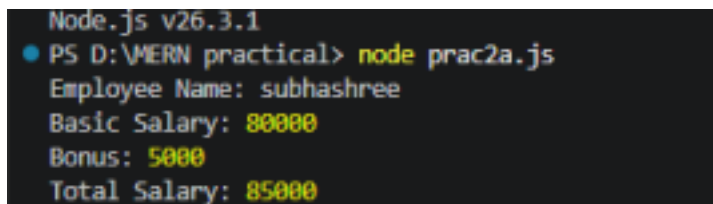
Roll No.:T007	Name:Subhashree jena
Class:TYIT	Batch:1
Date of Assignment:30/6/26	Date/Time of Submission:30/6/26

a. Create functions and call them with parameters.

CODE:-

```
function calculateSalary(name , basicSalary) {  
    let bonus = 5000;  
    let totalSalary = basicSalary + bonus;  
    console.log("Employee Name:" ,name);  
    console.log("Basic Salary:" ,basicSalary);  
    console.log("Bonus:" ,bonus);  
    console.log("Total Salary:" ,totalSalary);  
}  
calculateSalary("subhashree",80000);
```

OUTPUT:-



```
Node.js v26.3.1  
PS D:\MERN practical> node prac2a.js  
Employee Name: subhashree  
Basic Salary: 80000  
Bonus: 5000  
Total Salary: 85000
```

CODE:-

```
function generateBill(product , quantity,price) {  
    let Total = quantity*price;  
    let bill =Total +(Total *0.18);  
    console.log("product:" ,product);
```

```
console.log("Quantity:" ,quantity);

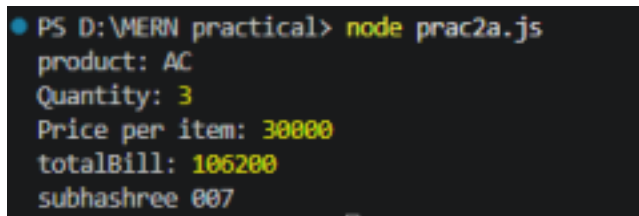
console.log("Price per item:" ,price);
console.log("totalBill:" ,bill);

}

generateBill("AC",3,30000);

console.log("subhashree 007")
```

OUTPUT:-



```
PS D:\MERN practical> node prac2a.js
product: AC
Quantity: 3
Price per item: 30000
totalBill: 106200
subhashree 007
```

b. Work with arrays and objects in JavaScript.

CODE:-

```
let Fruits=["Mango","Banana","Apple","Cherry"];

let student={

  name:"subhashree",

  age:20,

  city:"kolkata"

};

console.log("Fruits:",Fruits);

console.log("Second Fruits:",Fruits[1]);

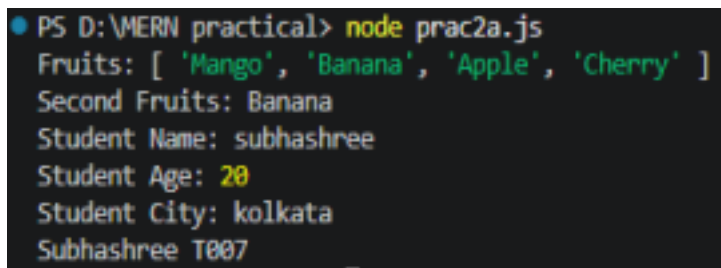
console.log("Student Name:",student.name);

console.log("Student Age:",student.age);

console.log("Student City:",student.city);

console.log("Subhashree T007");
```

OUTPUT:-

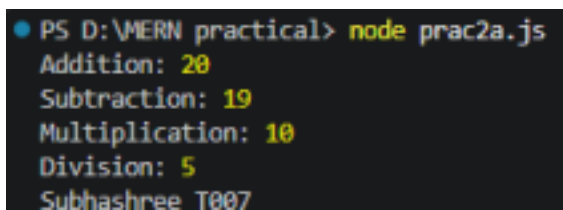


```
PS D:\MERN practical> node prac2a.js
Fruits: [ 'Mango', 'Banana', 'Apple', 'Cherry' ]
Second Fruits: Banana
Student Name: subhashree
Student Age: 20
Student City: kolkata
Subhashree T007
```

c. Demonstrate use of arrow functions.

CODE:-

```
let add=(a,b)=>{
    return a+b
};
let sub=(a,b)=>{
    return a-b
};
let mul=(a,b)=>{
    return a*b
};
let div=(a,b)=>{
    return a/b
};
console.log("Addition:",add(10,10));
console.log("Subtraction:",(22,19));
console.log("Multiplication:",(10,10));
console.log("Division:",(15,5));
console.log("Subhashree T007");
```

OUTPUT:-

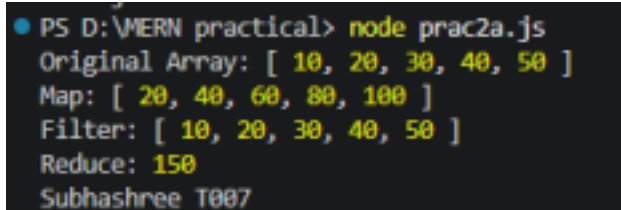
d. Write a

program using array methods (map, filter, reduce). CODE:-

```
let numbers=[10,20,30,40,50];
//map
let double=numbers.map(num =>num*2);
//filter
let greater25=numbers.filter(num=>25);
//reduce
let total=numbers.reduce((sum,num)=>sum+num,0);
console.log("Original Array:",numbers);
console.log("Map:",double);
```

```
console.log("Filter:",greater25);
console.log("Reduce:",total);
console.log("Subhashree T007");
```

OUTPUT:-



```
PS D:\MERN practical> node prac2a.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 10, 20, 30, 40, 50 ]
Reduce: 150
Subhashree T007
```

e. Create a

program that manipulates objects and displays output. CODE:-

```
//Creating an object
let student={
  rollNo:07,
  name:"subhashree",
  marks:92
};
console.log("Original Object");
console.log(student);
//2.Accessing Properties
console.log("\nAccessing Properties:");
console.log("Name:",student.name);
console.log("Marks:",student.marks);
//updating properties
student.marks=90;
console.log("\nAfter Updating Marks:");
console.log(student);
//Adding a new Property
student.marks=90;
console.log("\nAfter updating marks:");
console.log(student);
//Deleting a Property
delete student.rollNo;
console.log("\nAfter Deleting Roll Number:");
console.log(student);
```

OUTPUT:-

```
PS D:\MERN practical> node prac2a.js
Original Object
{ rollNo: 7, name: 'subhashree', marks: 92 }

Accessing Properties:
Name: subhashree
Marks: 92

After Updating Marks:
{ rollNo: 7, name: 'subhashree', marks: 90 }

After updating marks:
{ rollNo: 7, name: 'subhashree', marks: 90 }

After Deleting Roll Number:
{ name: 'subhashree', marks: 90 }
```

f. Implement a

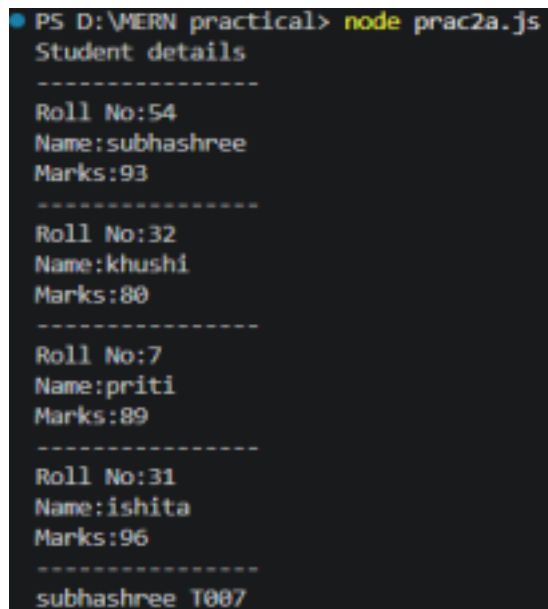
small JavaScript program combining functions, arrays, and objects. CODE:-

```
let students=[
  {
    rollNo:54,
    name:"subhashree",
    marks:93
  },
  {
    rollNo:32,
    name:"khushi",
    marks:80
  },
  {
    rollNo:7,
    name:"priti",
    marks:89
  },
  {
    rollNo:31,
    name:"ishita",
    marks:96
  }
];
```

```
//function display
function display(s)
{
console.log("Student details");
console.log("-----");
for(let stud of s)
{
    console.log(`Roll No:${stud.rollNo}`);
    console.log(`Name:${stud.name}`);
    console.log(`Marks:${stud.marks}`);
    console.log("-----");
}
}

//function call
display(students);
console.log("subhashree T007");
```

OUTPUT:-



```
PS D:\MERN practical> node prac2a.js
Student details
-----
Roll No:54
Name:subhashree
Marks:93
-----
Roll No:32
Name:khushi
Marks:80
-----
Roll No:7
Name:priti
Marks:89
-----
Roll No:31
Name:ishita
Marks:96
-----
subhashree T007
```